**A**

**FIELD PROJECT REPORT**

**ON**

# “Patient Waiting Time Analysis And Optimization In Hospital Using Business Analytics”

SUBMITTED TO

“SAVITRIBAI PHULE PUNE UNIVERSITY”

IN PARITAL FULFILLMENT OF THE REQUIREMENT

FOR THE AWARD OF THE DEGREE OF

MASTER OF BUSINESS ADMINISTRATION (MBA).

**UNDER THE GUIDENCE OF**

DR. RADHIKA KHAIRNAR

**SUBMITTED BY**

MR.PRANAV UMAKANT BIRADAR

MBA - I

**THROUGH**

उद्योगः कर्मसु कौशलम्

Dr. Moonje Institute of Management and Computer Studies

DR. MOONJE INSTITUTE OF MANAGEMENT AND

COMPUTER STUDIES, NASHIK

**(2025-2026)**

# DECLARATION

I undersigned hereby declare that, the field project titled **“Patient Waiting Time Analysis And Optimization In Hospital Using Business Analytics**” is executed as per the course requirement of a two-year full-time MBA Program of Savitribai Phule Pune University.

This report has not been submitted by me or any other person to any other University or Institution for a degree or diploma course. This is my own and original work.

**Place:** Nashik

**Mr. PRANAV .U. BIRADAR**
MBA-I (Business Analytics)

**Date:**

# ACKNOWLEDGEMENT

It is my pleasure to thank all who helped us directly or indirectly in the preparation of this project report. It is a great privilege to record our deep sense of gratitude to all the faculties who stood by us throughout the making of this field project report. It was very exciting for us to work on the project **"*Patient Waiting Time Analysis And Optimization Using Business Analytics*"**

During this work, I am gaining both practical as well as theoretical knowledge regarding it. My first thanks go to Dr. Moonje Institute of Management and Computer Studies, Nashik for providing us with an excellent environment like computer centre, library as well as providing us excellent faculty guiding for the completion of my field project, I am greatly obliged to my field project guide DR. Radhika Khairnar for her benevolent guidance, suggestions, and help for preparing this project.

**Place:** Nashik

**Mr. PRANAV .U. BIRADAR**
MBA-I (Business Analytics)

**Date:**

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# Dr. Moonje Institute

**of Management & Computer Studies, Nashik**

(Affiliated to S.P. Pune University & Approved by AICTE New Delhi)

(Accredited by NAAC with B++ Grade)

PUN Code : IMMN017930, DTE Code : 5119, Exam Code : 0688

# CERTIFICATE

**This is to certify that___________________________________**

Has satisfactorily completed his /her Field Project on the topic

______________________________________________________

______________________________________________________

As a partial fulfillment for the requirement of MBA degree A.Y. 2025-2026.

**Project Guide** **Academic Coordinator** **Director**

**Examiner/s** 1.____________________ 2.____________________

# INDEX

## EXECUTIVE SUMMARY

Healthcare service quality is one of the most important factors affecting patient satisfaction and hospital performance. Among different operational challenges faced by hospitals, patient waiting time is considered one of the major issues affecting overall healthcare experience. Long waiting time not only creates dissatisfaction among patients but also reduces operational efficiency and increases workload pressure on hospital staff.

This project titled "Patient Waiting Time Analysis and Optimization in Hospital Using Business Analytics" was conducted at DR. KARAD HOSPITAL, Satpur with the objective of analyzing hospital waiting time and identifying operational inefficiencies affecting patient flow. The study uses business analytics techniques to understand patient waiting patterns and suggest optimization strategies for improving hospital services.

The project mainly focuses on hospital processes such as registration, consultation, and billing departments where patients experience delays. Data for the study was collected through observations, patient feedback, hospital records, and interaction with hospital staff. The collected data was analyzed using descriptive and analytical methods supported by dashboard visualization.

The analysis showed that consultation departments experience the highest waiting time due to heavy patient load, limited staff availability, and inefficient scheduling processes. Registration delays were also identified because of manual documentation systems and crowd management issues.

The study suggests the implementation of digital queue management systems, online registration, optimized appointment scheduling, and real-time analytics dashboards for reducing patient waiting time and improving operational efficiency.

The healthcare industry is rapidly adopting modern technologies and data-driven systems to improve operational performance and patient care quality. In hospitals, patient waiting time has become a major concern because it directly affects patient satisfaction, hospital reputation, and service efficiency. Therefore, analyzing and optimizing patient flow has become essential for healthcare organizations.This project focused on studying operational inefficiencies responsible for increasing patient waiting time at DR. KARAD HOSPITAL, Satpur. The study examined different hospital departments including registration, consultation, billing, and crowd management systems to understand the causes of delays experienced by patients.

The research identified that consultation departments faced the highest operational pressure because of increasing patient volume and limited doctor availability during peak hours. Similarly, registration counters also experienced delays due to manual documentation processes and inefficient queue handling systems. These operational inefficiencies created congestion in hospital premises and reduced overall service efficiency.

The project used business analytics techniques for collecting, analyzing, and interpreting patient-related operational data. Information was collected through patient feedback, direct observation, staff interaction, and hospital records. The collected data was analyzed using Microsoft Excel, charts, graphs, and dashboard analysis techniques for better interpretation of patient flow and waiting time patterns.

The findings of the study indicate that staff availability has a major impact on patient waiting time. During busy hours, insufficient staff availability increased consultation delays and operational workload. Patients also highlighted the importance of effective communication and organized service systems for improving their hospital experience.

Another important observation of the study was the growing support for digital healthcare systems. A large number of respondents supported digital queue management systems and online appointment scheduling because such systems can reduce overcrowding, improve patient movement, and minimize unnecessary waiting periods.

The study also emphasized the importance of appointment scheduling systems in hospitals. Proper appointment management helps distribute patient flow evenly throughout the day and reduces pressure during peak operational hours. Appointment systems can improve doctor scheduling efficiency and support better healthcare service delivery.

Business analytics played an important role in identifying operational bottlenecks and understanding hospital workflow patterns. Dashboard visualization and data analysis helped in recognizing delays in registration, consultation, and billing processes. Through analytics-based insights, hospitals can improve decision-making and implement effective operational strategies.

The project further highlighted that hospital operational efficiency is directly connected with patient satisfaction. Patients who received organized and faster services expressed higher satisfaction levels, while long waiting periods created frustration and inconvenience. Therefore, reducing waiting time is necessary not only for operational improvement but also for enhancing healthcare quality and patient trust.

The study recommends several optimization strategies for improving hospital operations. These include implementation of digital registration systems, online appointment booking facilities, digital queue management systems, workforce planning during busy hours, improved coordination between departments, and real-time waiting time monitoring systems.

The project also suggests that hospitals should continuously monitor operational performance using business analytics dashboards and reporting systems. Data-driven healthcare management can help hospitals improve patient flow, reduce delays, optimize resources, and increase operational productivity.

Overall, this project concludes that technology adoption, efficient operational planning, and business analytics can significantly improve hospital performance and patient satisfaction. The implementation of digital healthcare systems and process optimization strategies can help DR. KARAD HOSPITAL, Satpur deliver faster, more organized, and patient-friendly healthcare services in the future.

# CHAPTER NO .1

# INTRODUCTION

# INTRODUCTION

Hospitals play a very important role in society by providing healthcare services to patients. In recent years, healthcare demand has increased significantly because of population growth, lifestyle diseases, and rising health awareness. As the number of patients visiting hospitals increases, managing hospital operations efficiently becomes a major challenge. One of the most common problems faced by hospitals is excessive patient waiting time.

Patient waiting time refers to the amount of time a patient spends waiting before receiving healthcare services. This includes waiting during registration, consultation, diagnostic testing, treatment, pharmacy services, and billing processes. Long waiting times negatively affect patient satisfaction and reduce the quality of healthcare services provided by hospitals.

In many hospitals, patients spend several hours waiting because of operational inefficiencies such as manual registration systems, poor appointment scheduling, insufficient staff availability, and ineffective queue management systems. These inefficiencies create stress for both patients and hospital employees. Delays also increase crowding in hospitals and reduce operational productivity.

Today, hospitals are increasingly using technology and business analytics to improve operational performance. Business analytics is the process of collecting, analyzing, and interpreting data to support decision-making and improve organizational efficiency. In hospitals, business analytics helps management identify delays, analyze patient flow, optimize resource allocation, and improve service quality.

This project focuses on analyzing patient waiting time at dr. Karad hospital, satpur using business analytics techniques. The study aims to identify operational factors causing delays and recommend optimization strategies for improving patient flow and reducing waiting time.

The project uses dashboard analysis, patient data interpretation, and operational observations to understand hospital workflow patterns. The recommendations provided in this study can support hospital management in improving healthcare service delivery and increasing patient satisfaction.

# STATEMENT OF THE PROBLEM

Healthcare organizations are expected to provide fast, efficient, and quality services to patients. However, many hospitals face operational challenges related to patient waiting time. Long waiting time has become one of the most common problems in hospital management and directly affects patient satisfaction, healthcare quality, and hospital efficiency.

At DR. KARAD HOSPITAL, Satpur**,** patients often experience delays during different stages of hospital service processes such as registration, consultation, diagnostic services, pharmacy, and billing. These delays increase patient frustration and create operational pressure on hospital staff.

One of the major reasons for long waiting time is the increasing number of patients visiting hospitals every day. During peak hours, hospital departments become overcrowded, resulting in delays in service delivery. Patients are required to wait for long periods before meeting doctors, completing documentation, or making payments.

Another important problem is the presence of manual operational systems. In many hospital departments, patient registration and billing activities are performed manually, which consumes additional time and increases the possibility of errors. Manual processes reduce workflow efficiency and create bottlenecks in patient movement.

Improper appointment scheduling is also an important issue affecting hospital operations. Lack of effective scheduling systems leads to uneven patient distribution throughout the day. As a result, hospitals experience excessive patient crowding during certain hours while resources remain underutilized during other periods.

Staff availability is another major operational concern. Limited availability of doctors, nurses, and administrative staff during busy hours increases patient waiting time. Insufficient staffing creates delays in consultation and administrative procedures, reducing overall operational productivity.

Poor coordination between hospital departments further contributes to inefficiencies in patient flow management. Delays in communication and process handling between registration, consultation, laboratory, and billing departments create unnecessary waiting periods for patients.

Therefore, there is a strong need to analyze patient waiting time and identify operational inefficiencies affecting healthcare service delivery at DR. KARAD HOSPITAL, Satpur. The study aims to use business analytics techniques to understand waiting time patterns, evaluate operational problems, and recommend optimization strategies for improving patient flow and reducing delays in hospital services.

# OBJECTIVE OF THE PROJECT

The main objective of this study is to analyze patient waiting time and identify operational inefficiencies in hospital services using business analytics techniques. The study focuses on understanding patient flow patterns and healthcare service delays at DR. KARAD HOSPITAL, Satpur. It aims to examine operational problems in registration, consultation, billing, and crowd management processes.

The project also studies the impact of staff availability and appointment scheduling on patient waiting time. Business analytics tools are used to analyze hospital workflow and identify bottlenecks affecting service efficiency. The study focuses on improving patient satisfaction through better operational management and faster healthcare services.

It also aims to recommend optimization strategies for reducing waiting time and improving patient flow. The project emphasizes the importance of digital systems such as online appointment scheduling and digital queue management. The study supports data-driven decision-making for improving hospital operations and service quality.

Overall, the project aims to enhance healthcare efficiency, patient experience, and operational productivity through process optimization and technology adoption.

The specific objectives of the study are as follows:

1. To analyze waiting time in different hospital departments.
2. To identify operational factors causing delays in hospital services.
3. To recommend optimization strategies using business analytics.
4. To study patient satisfaction regarding hospital waiting time.

# THEROTICAL FRAMEWORK

The theoretical framework of this study is based on the concepts of Business Analytics, Hospital Operations Management, Queue Management Theory, and Patient Flow Management. These concepts help in understanding the causes of patient waiting time and identifying suitable optimization strategies for improving healthcare services.

Hospitals are service organizations where operational efficiency is very important for providing quality patient care. In healthcare systems, patient waiting time is considered one of the major indicators of service quality and operational performance. Long waiting times reduce patient satisfaction and negatively affect hospital productivity.

This study uses business analytics as a framework for analyzing operational inefficiencies affecting patient flow, Business analytics involves collecting, analyzing, and interpreting data to support better decision-making and operational improvements. In hospitals, analytics can help management identify bottlenecks, monitor patient flow, and optimize healthcare processes.

The study is also supported by the concept of Queue Management Theory. Queue management theory explains how waiting lines are formed when service demand exceeds service capacity. In hospitals, queues are created during registration, consultation, laboratory testing, pharmacy services, and billing processes. Inefficient queue systems increase waiting time and reduce service quality. Proper queue management helps reduce congestion and improve patient movement within the hospital.

- ## Concept of Business Analytics

Business Analytics refers to the process of collecting, analyzing, interpreting, and visualizing data to support better decision-making and improve organizational performance. In hospitals, business analytics helps management understand patient flow, operational efficiency, waiting time patterns, and service-related problems.

Business analytics uses data analysis tools, charts, dashboards, and statistical techniques to identify operational bottlenecks and suggest suitable optimization strategies. In healthcare organizations, analytics supports faster decision-making and improves healthcare service quality.

## • Analysis Conducted in the Study

This study mainly focuses on analyzing patient waiting time and operational inefficiencies at DR. KARAD HOSPITAL, Satpur. Different types of operational analysis were conducted to understand patient flow and healthcare service delays.

The following analysis was conducted in the project:

- Patient Waiting Time Analysis
- Registration Delay Analysis
- Staff Availability Impact Analysis
- Patient Satisfaction Analysis
- Crowd Management Analysis
- Appointment System Analysis
- Digital Queue System Support Analysis
- Department-wise Operational Efficiency Analysis

These analyses helped identify the major operational factors affecting hospital efficiency and patient satisfaction.

## • Importance of Data Analysis in Hospitals

Data analysis helps hospitals convert raw operational data into useful information for improving healthcare services. Through proper data interpretation, hospitals can identify peak hours, staff workload, patient congestion, and process inefficiencies.

Data analysis in hospitals helps in:

- Reducing patient waiting time
- Improving patient satisfaction
- Enhancing operational efficiency
- Optimizing staff utilization
- Improving workflow management
- Supporting better healthcare planning

- **Technology Adoption in Healthcare**

Modern hospitals are increasingly adopting digital technologies for improving operational performance and healthcare quality. Technologies such as digital registration systems, online appointment scheduling, electronic medical records, and queue management systems help reduce delays and improve patient convenience.

The study also considers the relationship between operational factors and patient waiting time. The major operational factors influencing waiting time include:

- High patient volume
- Manual registration systems
- Limited staff availability
- Inefficient appointment scheduling
- Lack of digital systems
- Poor coordination between departments
- Crowd management issues

The theoretical framework assumes that improvements in operational efficiency and technology adoption can significantly reduce patient waiting time. Business analytics tools such as dashboard analysis, data visualization, and process analysis can help hospitals identify delays and implement effective solutions.

## SIGNIFICANCE OF THE PROJECT

This study is important because patient waiting time is one of the major operational problems faced by hospitals. Long waiting periods negatively affect patient satisfaction, healthcare quality, and overall hospital efficiency. Therefore, analyzing and reducing patient waiting time has become essential for improving healthcare service delivery.

The study conducted at DR. KARAD HOSPITAL, Satpur helps identify operational inefficiencies affecting patient flow and suggests suitable optimization strategies using business analytics. The findings of this study can provide practical solutions for improving hospital operations and enhancing patient experience.

The significance of this study is explained below:

### Importance for Hospital Management

The study helps hospital management understand the major causes of patient waiting time and operational delays. Through business analytics and dashboard analysis, management can identify bottlenecks in hospital processes such as registration, consultation, and billing.

The recommendations provided in this study can help management:

- Improve operational efficiency
- Reduce patient waiting time
- Optimize resource utilization
- Improve workflow management
- Enhance service quality

The study also supports data-driven decision-making for hospital administration.

**Importance for Patients**

Patients are the most important stakeholders in healthcare services. Long waiting time creates frustration, dissatisfaction, and inconvenience for patients.

This study is significant for patients because it focuses on improving patient flow and reducing delays in hospital services. Better operational systems can improve patient experience and increase satisfaction levels.

The implementation of optimization strategies suggested in this study can help patients receive faster and more efficient healthcare services

**Importance for Hospital Staff**

Hospital staff members such as doctors, nurses, and administrative employees often experience workload pressure due to inefficient operational systems and overcrowding.

This study helps identify workflow problems affecting staff productivity and operational performance. Improved scheduling systems, digital registration processes, and queue management techniques can reduce work pressure on hospital employees and improve coordination between departments.

**Importance for Researchers and Students**

This study provides useful information for future researchers and students interested in healthcare analytics, hospital management, and business analytics applications in healthcare systems.

The project can serve as a reference for future studies related to:

- Patient waiting time analysis
- Hospital process optimization
- Queue management systems
- Healthcare analytics
- Operational efficiency in hospitals

The study also helps students understand the practical application of business analytics in healthcare organizations

# CHAPTER NO .2

# LITERATURE REVIEW

# LITERATURE REVIEW

Review of Literature is an important part of any research project. It helps the researcher understand previous studies, theories, concepts, and findings related to the selected topic. In this project, the literature review focuses on patient waiting time analysis, hospital service quality, queue management, patient satisfaction, and the use of business analytics in healthcare operations.

Hospitals are service-based organizations where timely service delivery plays an important role in patient satisfaction. Many researchers have observed that long waiting time is one of the major causes of dissatisfaction among patients. Waiting time affects not only patient experience but also hospital reputation and operational efficiency.

Business analytics has become an important tool in healthcare management. It helps hospitals collect and analyze data related to patient flow, staff availability, appointment scheduling, registration time, consultation time, and billing delays. Through proper data analysis, hospitals can identify bottlenecks and improve service delivery.

This chapter reviews various studies and concepts related to patient waiting time and hospital process optimization.

- **Concept of Patient Waiting Time**

Patient waiting time refers to the time spent by a patient before receiving required healthcare services. It begins from the moment a patient enters the hospital and continues until the patient receives consultation, treatment, testing, billing, or discharge services.

Waiting time may occur at different hospital stages such as:

- Registration counter
- Doctor consultation
- Laboratory or diagnostic services
- Pharmacy
- Billing counter
- Report collection
- Follow-up appointment

Long waiting time creates inconvenience for patients and increases crowding in hospitals. It also affects patient satisfaction and service quality. Therefore, patient waiting time is considered an important indicator of hospital performance.

In hospital operations, waiting time occurs when patient demand is higher than service capacity. For example, if many patients arrive at the same time but only one registration counter is available, patients must wait in a queue. Similarly, if doctor availability is limited, consultation waiting time increases.

- **Concept of Hospital Operations Management**

Hospital operations management refers to planning, organizing, and controlling hospital activities to provide efficient healthcare services. It includes managing doctors, staff, patients, equipment, departments, appointments, records, and hospital processes.

Effective hospital operations management helps in:

- Reducing patient waiting time
- Improving staff productivity
- Managing patient flow
- Improving service quality
- Reducing overcrowding
- Improving patient satisfaction

Poor hospital operations management creates delays and inefficiencies. If registration, consultation, and billing processes are not properly coordinated, patients experience long waiting time. Therefore, hospitals need systematic operational planning and process improvement methods.

- **Patient Waiting Time and Patient Satisfaction**

Patient satisfaction means the level of happiness or approval patients feel after receiving hospital services. It depends on factors such as:

- Waiting time
- Staff communication
- Doctor availability
- Cleanliness

- Billing process
- Treatment quality
- Hospital facilities
- Patient guidance

Among these factors, waiting time has a strong impact on patient satisfaction. If patients wait for a long time without proper communication, they may become frustrated and dissatisfied.

In this project, the dashboard showed that patient satisfaction with waiting time is an important concern. Some patients were satisfied, but many patients supported improvement in hospital queue and process management.

Therefore, reducing waiting time is essential for improving patient satisfaction at DR. KARAD HOSPITAL, Satpur.

- **<u>Business Analytics in Healthcare</u>**

Business Analytics is the process of using data, statistical methods, and technology to support decision-making. In healthcare, business analytics is used to improve hospital operations and patient care.

Healthcare organizations generate large amounts of data every day, including patient records, appointment data, billing data, staff data, and department-wise service time. Business analytics helps convert this data into useful insights.

Business analytics in hospitals can help in:

- Analyzing patient waiting time
- Identifying peak hours
- Predicting patient flow
- Monitoring staff performance
- Improving appointment scheduling
- Reducing operational delays
- Improving patient satisfaction

For this project, business analytics is useful because it helps study waiting time patterns and identify process inefficiencies at DR. KARAD HOSPITAL, Satpur.

# CHAPTER NO .3

# RESEARCH METHODOLOGY

## 1. Introduction

Research Methodology refers to the systematic methods and techniques used for collecting, analyzing, and interpreting data for a research study. It explains how the research is conducted and how information is gathered to achieve the objectives of the study.

In this project, research methodology was used to analyze patient waiting time and identify operational inefficiencies at DR. KARAD HOSPITAL, Satpur. The study focuses on understanding patient flow, waiting time patterns, staff availability, registration delays, and queue management issues using business analytics techniques.

This chapter explains the research design, data collection methods, sampling techniques, analytical tools, and limitations of the study.

## 2. Research Design

Research Design refers to the overall framework or plan used for conducting the study. It helps in systematic collection and analysis of data.

For this project, a descriptive and analytical research design was used because the study focuses on:

- Observing patient waiting time patterns
- Analyzing hospital operational processes
- Identifying causes of delays
- Evaluating patient satisfaction
- Suggesting optimization strategies

The research design helped in collecting structured information related to patient flow and operational efficiency at DR. KARAD HOSPITAL.

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## 3. Data Collection Methods

- **Primary Data Collection**

  The study is based on both primary and secondary data.

Primary Data refers to original data collected directly by the researcher for the specific purpose of the study.

For this project, primary data was collected through:

- **Observation**

The researcher observed patient movement and waiting time in different hospital departments such as registration, consultation, and billing.

- **Patient Feedback**

Information was collected from patients regarding their experience, satisfaction level, and waiting time issues.

- **Staff Interaction**

Discussions were conducted with hospital staff to understand operational challenges and workflow problems.

- **Hospital Records**

Basic operational records related to patient flow and service timing were used for analysis.

- **Secondary Data Collection**

Secondary Data refers to information already collected and published by other sources.

Secondary data for this study was collected from:

- Research journals
- Healthcare reports
- Books
- Websites
- Articles related to hospital management
- Previous research papers on waiting time analysis and healthcare analytics

Secondary data helped in understanding theoretical concepts and previous research findings related to patient waiting time and business analytics.

## Sampling Method

Sampling refers to selecting a group of respondents from the total population for data collection and analysis.

For this study, a Convenience Sampling Method was used.

Convenience sampling means selecting respondents who are easily available and willing to participate in the study. Patients visiting the hospital during the data collection period were selected as respondents.

This method was chosen because of time limitations and ease of data collection.

### <u>Sample Size</u>

The sample size for the study consisted of 82 respondents from DR. KARAD HOSPITAL, Satpur..

## Study Area

The study was conducted at DR. KARAD HOSPITAL, Satpur.

The project focuses on patient waiting time and operational efficiency in hospital departments such as:

- Registration Department
- Consultation Department
- Billing Department

The study also examines patient flow, crowd management, and service delivery processes within the hospital

## Data Collection Tool

A structured questionnaire was used as the primary data collection tool. The questionnaire included multiple-choice questions related to consumer behavior, platform preference, and customer satisfaction.

## Tools Used for Analysis

The collected data was analyzed using Microsoft Excel. Various tools such as percentage analysis, pie charts, bar graphs, and column charts were used to interpret the data.

## Data Type

The study is based on quantitative data, as it involves numerical responses collected through the questionnaire.

# CHAPTER NO .4

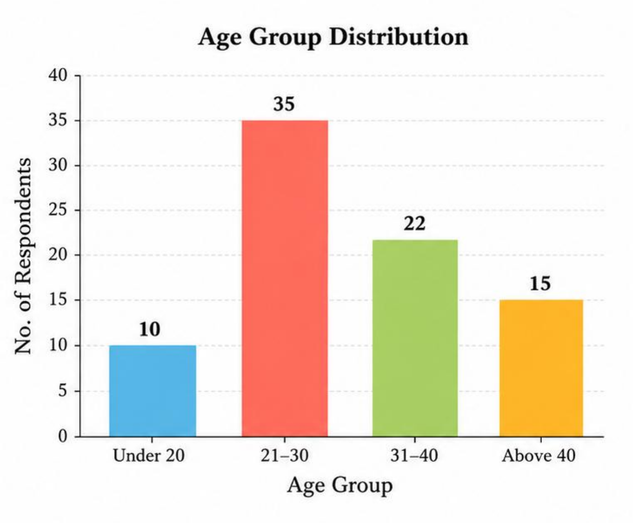
# Data Analysis and Interpretation

**Sec A:**

## 4.1Age-wise Distribution of Respondents

***Table 4.1**: Age-wise Distribution of Respondents*

| Age Group | No. of Respondents | Percentage |
|---|---|---|
| Below 20 Years | 10 | 12.2% |
| 21 – 30 Years | 35 | 42.7% |
| 31 – 40 Years | 22 | 26.8% |
| Above 40 Years | 15 | 18.3% |
| Total | 82 | 100% |




### Observation

- Majority of respondents belong to the age group of 20–30 years.
- Respondents from the 30–40 years category are moderate in number.
- Fewer respondents belong to below 20 years and above 40 years age groups.
- Young adults visit the hospital more frequently compared to other age groups.

### Interpretation

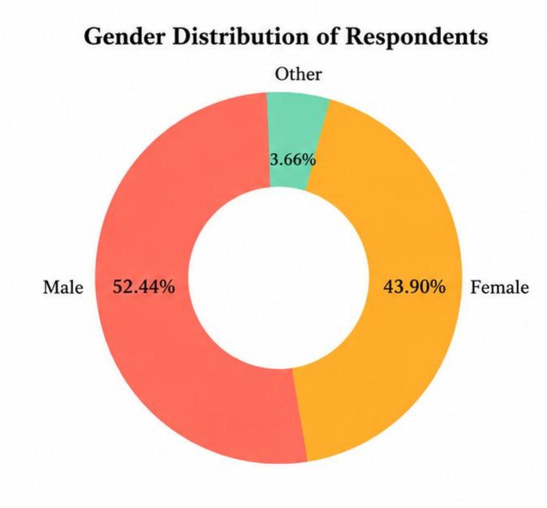
The analysis indicates that young adults form the largest patient group visiting DR. KARAD HOSPITAL, Satpur. This age group generally expects quick and efficient healthcare services because of busy work schedules and active lifestyles.

The high number of young adult patients increases pressure on hospital operations during peak hours, especially in registration and consultation departments. The hospital should therefore focus on better appointment scheduling, digital queue systems, and faster patient handling processes to manage patient flow efficientl

## 4.2Gender-wise Distribution of Respondents

***Table 4.2:*** *Gender Distribution of Respondents*

| *Gender* | *No. of Respondents* | *Percentage* |
|---|---|---|
| ***Male*** | 43 | 52.44% |
| ***Female*** | 36 | 43.90% |
| ***Other*** | 3 | 3.66% |
| ***Total*** | 82 | 100% |




### Observation

- Male respondents are slightly higher than female respondents.
- Both male and female patients experience waiting time issues.
- Hospital services are utilized almost equally by different gender groups.
- Waiting time problems are common among all patients regardless of gender.

### Interpretation

The findings show that patient waiting time is a common operational issue affecting all gender groups equally. Since both male and female respondents reported delays during hospital visits, hospital management must focus on improving service efficiency for all patients.

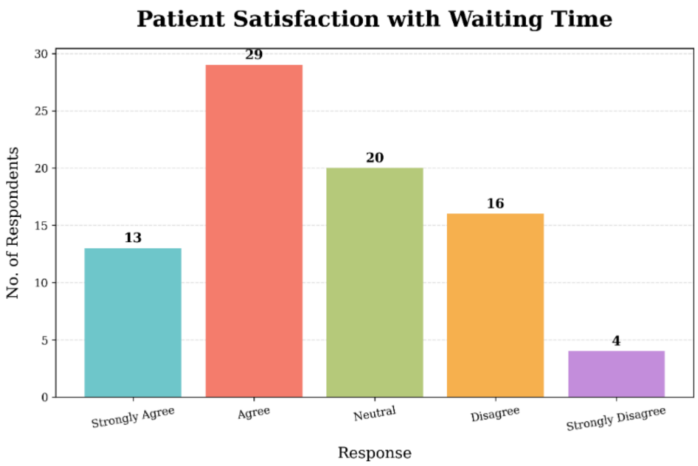
Improved queue management, faster registration procedures, and better coordination between departments can help reduce delays and improve patient satisfaction. The hospital should ensure equal and efficient service delivery to all categories of patients.

## Sec B:

## 4.3Patient Satisfaction with Waiting Time

***Table 4.3:*** *Satisfaction Level of Patients*

| Response | No. of Respondents | Percentage |
|---|---|---|
| Strongly Agree | 13 | 16% |
| Agree | 29 | 35% |
| Neutral | 20 | 24% |
| Disagree | 16 | 19% |
| Strongly Disagree | 4 | 5% |
| Total | 82 | 100 |




### Observation

- Most respondents are satisfied with the hospital waiting time.
- A moderate number of respondents gave neutral responses.
- Some respondents are dissatisfied, indicating delays in services.
- Overall, the results show a need to improve operational efficiency and reduce waiting time.

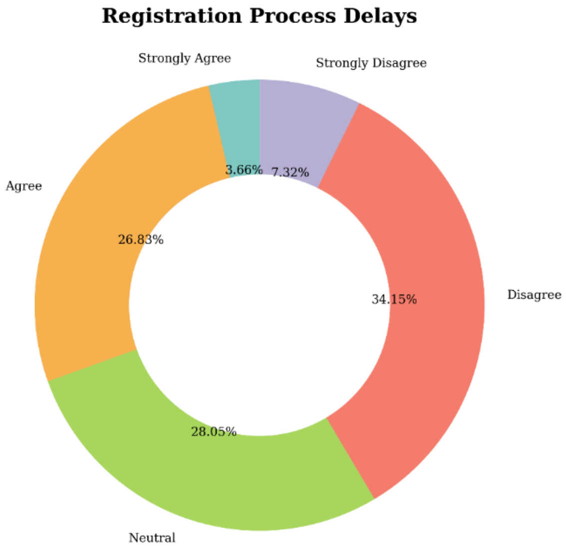
### Interpretation

The majority of respondents show a positive opinion toward the hospital's waiting time, as more participants agreed than disagreed. However, a considerable number of respondents expressed dissatisfaction, indicating that waiting time is still an issue for many patients. This suggests that the hospital should focus on improving operational efficiency and reducing patient waiting time to increase overall satisfaction.

## 4.4 Registration Delay Analysis

***Table 4.4:*** *Registration Process Delay*

| Response | No. of Respondents | Percentage |
|---|---|---|
| **Strongly Agree** | 3 | 3.66% |
| **Agree** | 22 | 26.83% |
| **Neutral** | 23 | 28.05% |
| **Disagree** | 28 | 34.15% |
| **Strongly Disagree** | 6 | 7.32% |
| **Total** | 82 | 100% |




### Observation

- Total respondents were **82**.
- Majority (**34.15%**) disagreed about registration process delays.
- **28.05%** respondents were neutral.
- **26.83%** agreed that delays exist.
- Overall, negative responses were higher than positive responses.

### Interpretation

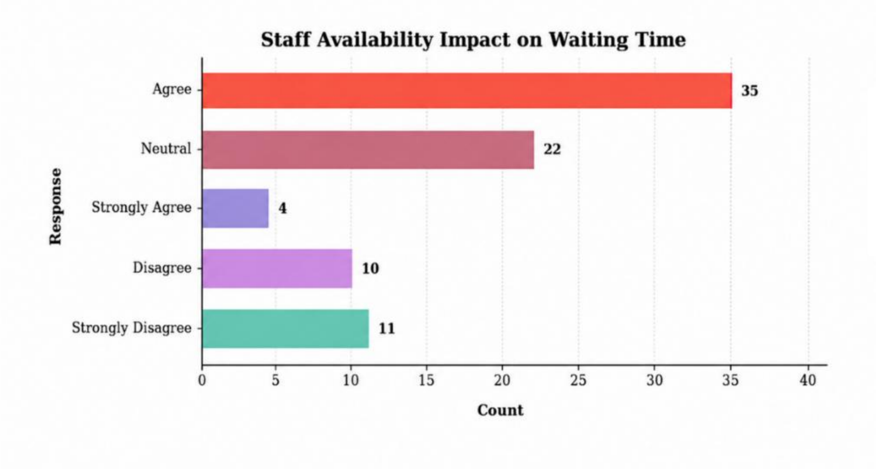
The analysis indicates that manual registration systems create operational inefficiencies and increase patient waiting time. Slow documentation processes and heavy patient load create congestion in registration departments.

## Sec C:

### 4.5Staff Availability Impact Analysis

***Table 4.5:*** *Staff Availability Impact on Waiting Time*

| Response | No. of Respondent | Percentage |
|---|---|---|
| **Strongly Agree** | 4 | 13% |
| **Agree** | 35 | 43% |
| **Neutral** | 22 | 27% |
| **Disagree** | 10 | 12% |
| **Strongly Disagree** | 11 | 5% |
| **Total** | 82 | 100% |




### Observation

- Most respondents believe staff availability affects waiting time.
- Delays increase during busy hours because of limited staff.
- Consultation waiting time is affected by doctor availability.
- Staff shortage creates operational pressure.

### Interpretation

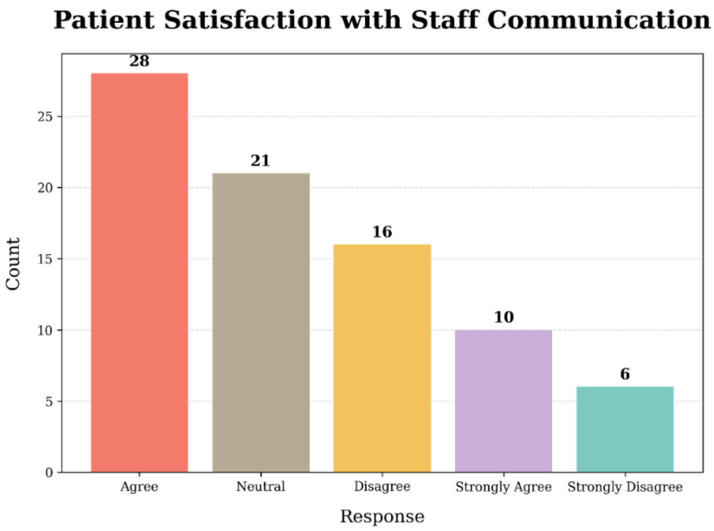
The findings show that insufficient staff availability is one of the major factors responsible for long patient waiting time. During peak hours, limited availability of doctors and administrative staff increases service delays and operational inefficiencies.

The hospital should improve workforce planning and increase staff availability during busy periods. Proper staff scheduling can help reduce waiting time and improve patient service efficiency.

## 4.6Patient Satisfaction with Staff Communication

***Table 4.6:*** *Patient Satisfaction with Staff Communication*

| Response | No. of Respondents | Percentage |
|---|---|---|
| **Strongly Agree** | 10 | 12% |
| **Agree** | 28 | 35% |
| **Neutral** | 21 | 26% |
| **Disagree** | 16 | 20% |
| **Strongly Disagree** | 6 | 7% |
| **Total** | 82 | 100% |




### Observation

- Majority of respondents are satisfied with staff communication.
- Some patients gave neutral responses regarding communication quality.
- A smaller number of respondents were dissatisfied with staff interaction.
- Positive communication improves patient experience during hospital visits.

### Interpretation

The analysis indicates that staff communication plays an important role in improving patient satisfaction and healthcare experience. Most respondents were satisfied with the communication and behavior of hospital staff, which reflects positive interaction between patients and employees.

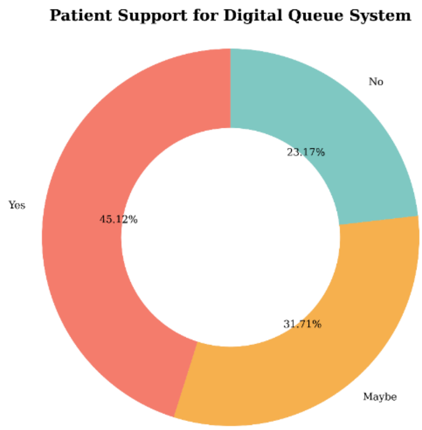
However, some patients were neutral or dissatisfied, which suggests that communication quality may vary during busy hours or high workload situations.

**Sec D :**

### 4.7Digital Queue System Support

***Table 4.7:*** *Support for Digital Queue System*

| Response | No. of Respondents | Percentage |
|---|---|---|
| **Yes** | 37 | 45.12% |
| **No** | 19 | 23.17% |
| **Maybe** | 26 | 31.71% |
| **Total** | 82 | 100% |




### Observation

- 45% respondents supported the implementation of a digital queue management system in the hospital.
- 33% respondents were uncertain and selected "Maybe" regarding digital queue systems.
- 23% respondents did not support the use of digital queue systems.
- Majority of patients believe that digital systems can help reduce waiting time and improve service efficiency.

### Interpretation

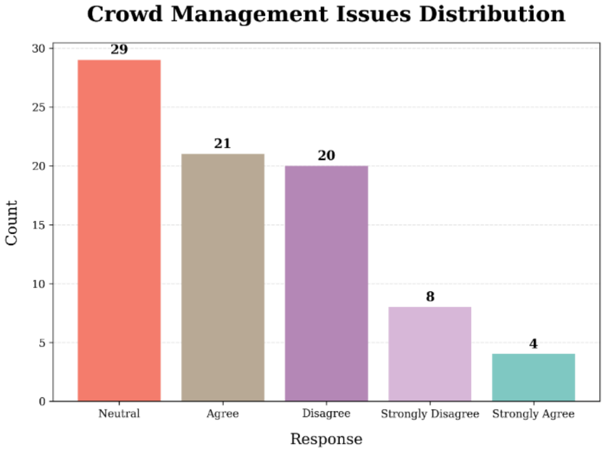
The analysis indicates that a large number of patients are interested in technology-based healthcare solutions for improving hospital services. The majority of respondents supported digital queue systems because they believe such systems can organize patient flow more effectively and reduce unnecessary waiting time.

Digital queue systems can improve operational efficiency by reducing overcrowding, avoiding confusion in waiting areas, and managing patient movement systematically.

## 4.8Crowd Management Issue Analysis.

***Table 4.8:*** *Crowd Management Issues.*

| Reason | No. of Respondents | Percentage |
|---|---|---|
| **Strongly Agree** | 4 | 5% |
| **Agree** | 21 | 26% |
| **Neutral** | 29 | 35% |
| **Disagree** | 20 | 24% |
| **Strongly Disagree** | 8 | 10% |
| **Total** | 82 | 100 |




### Observation

- Majority of respondents gave a neutral response regarding crowd management effectiveness.
- A considerable number of respondents agreed that crowd management issues exist in the hospital.
- Some respondents disagreed and strongly disagreed regarding crowd management problems.
- Only a small number of respondents strongly agreed about the effectiveness of crowd management.
- The responses indicate that crowd management requires improvement, especially during peak hours.

### Interpretation

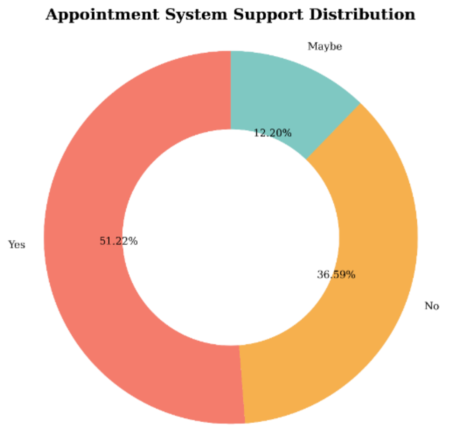
The analysis indicates that crowd management is one of the major operational challenges affecting hospital efficiency and patient satisfaction. A large number of respondents reported that overcrowding creates delays, confusion, and inconvenience during hospital visits.

Poor crowd management negatively affects patient flow and increases pressure on hospital staff, especially during busy hours. Congestion in registration, consultation, and billing areas leads to longer waiting times and reduces the overall quality of healthcare services.

## 4.9Distribution of Appointment System Support

***Table 4.9:*** *Distribution of Appointment System Support.*

| Response | No. of Respondents | Percentage |
|---|---|---|
| **Yes** | 42 | 51.22% |
| **No** | 30 | 36.59% |
| **Maybe** | 10 | 12.20% |
| **Others** | 82 | 100% |




### Observation

- Majority of respondents preferred the appointment system in the hospital.
- Some respondents gave neutral opinions regarding appointment scheduling.
- A smaller number of patients did not prefer appointment systems.
- Patients believe appointment systems can reduce waiting time and improve service management.
- Proper appointment scheduling helps avoid overcrowding during peak hours.

### Interpretation

The analysis indicates that most patients support the use of appointment systems because they help reduce unnecessary waiting time and improve patient flow management. Appointment scheduling allows hospitals to manage patient visits systematically and avoid overcrowding during peak hours.

An effective appointment system can improve operational efficiency by distributing patient load evenly throughout the day. It also helps doctors and hospital staff manage consultation schedules more efficiently. Patients benefit from reduced delays and better service experience.

# CHAPTER NO .5

# CONCLUSION, FINDINGS & SUGGESTIONS

# <u>CONCLUSION</u>

The study on **“Patient Waiting Time Analysis and Optimization in Hospital Using Business Analytics”** conducted at **DR. KARAD HOSPITAL, Satpur** focused on analyzing patient waiting time and identifying operational inefficiencies affecting healthcare services.

The analysis revealed that patient waiting time is influenced by several operational factors such as registration delays, staff availability, crowd management issues, and inefficient appointment scheduling. Consultation departments experienced the highest waiting time because of high patient load and limited staff availability during peak hours.

The study also showed that patient satisfaction is directly related to waiting time and service efficiency. Long waiting periods create dissatisfaction and inconvenience for patients. Most respondents supported the implementation of digital queue systems and improved appointment scheduling for better hospital management.

Business analytics and dashboard analysis helped identify operational bottlenecks and provided useful insights for improving patient flow. The findings indicate that technology-based healthcare solutions such as digital registration systems, queue management systems, and real-time monitoring can significantly reduce waiting time and improve operational efficiency.

Overall, the study concludes that proper operational planning, technology adoption, and efficient patient flow management are essential for improving healthcare service quality and patient satisfaction at DR. KARAD HOSPITAL, Satpur.

**1. Increase in Patient Waiting Time:**

The study found that patient waiting time is one of the major operational challenges at DR. KARAD HOSPITAL, Satpur. Patients experience delays mainly during registration, consultation, and billing processes, especially during peak hours. Long waiting periods negatively affect patient satisfaction and healthcare service quality.

**2. Operational Inefficiencies in Hospital Services:**

The study identified several operational inefficiencies such as manual registration systems, poor appointment scheduling, staff shortages, and crowd management issues. These factors increase patient congestion and create delays in healthcare service delivery.

**3. Impact of Staff Availability:**

The findings revealed that staff availability plays an important role in managing patient waiting time. Limited availability of doctors and administrative staff during busy hours increases operational pressure and affects patient flow efficiency.

**4. Importance of Patient Satisfaction:**

The study concluded that patient satisfaction is directly related to waiting time and service efficiency. Patients receiving faster and organized healthcare services were more satisfied, whereas long delays reduced overall patient experience.

**5. Support for Digital Queue Systems:**

A large number of respondents supported the implementation of digital queue management systems and online appointment scheduling. Patients believe that digital healthcare systems can reduce waiting time and improve operational efficiency.

**6. Crowd Management Challenges:**

The study found that overcrowding in waiting areas and poor crowd management significantly affect hospital operations. Congestion during peak hours increases confusion, delays, and inconvenience for patients and hospital staff.

**7. Role of Business Analytics in Healthcare:**

The use of business analytics and dashboard analysis helped identify operational bottlenecks and patient flow issues. Data analysis provided useful insights for improving hospital operations and supporting better decision-making.

**8. Need for Process Optimization:**

The study concludes that implementation of digital registration systems, queue management techniques, improved appointment scheduling, and proper workforce planning can significantly reduce patient waiting time and improve healthcare service quality at DR. KARAD HOSPITAL, Satpur.

## OBJECTIVE-WISE CONCLUSION

### Objective 1: To identify operational factors causing delays in hospital services

The study analyzed patient waiting time in various hospital departments such as Registration/Reception, Consultation, Billing, and Medicine Counter. The findings revealed that waiting time differs across departments depending on patient flow, staff availability, and service efficiency. The Consultation department recorded the highest waiting time due to increased patient volume and limited consultation duration.

The Registration/Reception and Billing departments showed moderate delays because of documentation and payment procedures. The Medicine Counter had comparatively lower waiting time, although delays were observed during peak hours. The study concludes that reducing waiting time through proper workflow management and digital systems can improve patient satisfaction and hospital service quality.

### Objective 2: To identify operational factors causing delays in hospital services

The study identified several operational factors responsible for increasing patient waiting time in the hospital. Major factors include manual registration systems, insufficient staff availability, poor appointment scheduling, overcrowding, and ineffective queue management.

The findings revealed that delays occur mainly during peak hours when patient volume increases significantly. Lack of digital systems and slow documentation processes further increase operational inefficiencies. The study concludes that proper operational planning and workflow improvements are necessary for reducing delays and improving service efficiency.

### Objective 3: To recommend optimization strategies using business analytics

The study found that business analytics plays an important role in improving hospital operations and reducing patient waiting time. Dashboard analysis and data interpretation helped identify bottlenecks and operational inefficiencies affecting patient flow.

The study recommends implementation of digital registration systems, queue management systems, online appointment scheduling, and real-time analytics dashboards for better operational management. These strategies can improve workflow efficiency, reduce overcrowding, and enhance patient satisfaction.

**Objective 4: To study patient satisfaction regarding hospital waiting time**

The study concluded that patient satisfaction is directly affected by waiting time and operational efficiency. Patients who received faster and organized services were more satisfied, whereas long waiting periods created dissatisfaction and frustration.

The findings also showed that staff communication and organized patient handling positively influence patient experience. Therefore, reducing waiting time and improving healthcare service management can significantly increase patient satisfaction and hospital reputation.

# FINDINGS

**1..Age Group of Respondents:**

- The majority of respondents belong to the 20–30 years age group.
- This indicates that young adults are the major users of hospital services at DR. KARAD HOSPITAL, Satpur.

**2.Gender Distribution:**

- Male respondents are slightly higher compared to female respondents.
- Both male and female patients experience waiting time issues during hospital visits.

**3.Patient Waiting Time:**

- Consultation departments experience the highest waiting time compared to registration and billing departments.
- Long waiting periods mainly occur during peak patient hours.

**4.Registration Process Delays:**

- Many respondents agreed that registration procedures are time-consuming.
- Manual documentation and paperwork increase operational delays and patient congestion.

**5.Staff Availability Impact:**

- Majority of respondents believe that staff availability directly affects patient waiting time.
- Limited availability of doctors and administrative staff increases delays during busy hours.

**6.Patient Satisfaction with Waiting Time:**

- More than half of the respondents are satisfied with hospital services.
- However, some patients are dissatisfied because of long consultation and billing delays.

**7.Patient Satisfaction with Staff Communication:**

- Most respondents are satisfied with staff communication and behavior.
- Positive communication improves patient experience and healthcare satisfaction.

**8.Support for Digital Queue System:**

- Majority of respondents support digital queue management systems in the hospital.
- Patients believe digital systems can reduce waiting time and improve service efficiency.

**9.Appointment System Preference:**

- Most respondents prefer appointment-based healthcare services.
- Patients believe appointment scheduling can reduce overcrowding and improve patient flow management.

**10.Crowd Management Issues:**

- Many respondents identified crowd management as a major operational issue.
- Overcrowding during peak hours increases waiting time and operational pressure.

**11.Department-wise Operational Efficiency:**

- Consultation departments experience more operational pressure compared to registration and billing sections.
- Workflow inefficiencies affect patient movement across hospital departments.

**12.Impact of Manual Systems:**

- Manual registration and documentation systems increase patient waiting time.
- Lack of digital systems affects operational efficiency and patient handling speed.

**13.Role of Business Analytics:**

- Business analytics and dashboard analysis helped identify waiting time patterns and operational bottlenecks.
- Data analysis supported better understanding of patient flow and service inefficiencies.

**14.Technology Adoption in Healthcare:**

- Patients show positive response toward technology-based healthcare solutions.
- Digital registration systems and queue management tools are expected to improve operational performance.

**15.Overall Finding:**

- Patient waiting time at DR. KARAD HOSPITAL is mainly influenced by operational inefficiencies, staff availability, and crowd management issues.
- Implementation of digital systems and improved operational planning can significantly improve patient satisfaction and healthcare service quality.

## SUGGESTIONS

1. **Implement digital registration systems**, to reduce manual paperwork and minimize patient waiting time at registration counters.
2. **Introduce digital queue management systems**, such as token systems and digital display boards, to improve patient flow and reduce overcrowding.
3. **Improve appointment scheduling systems**, to distribute patient visits properly throughout the day and avoid peak-hour congestion.
4. **Increase staff availability during busy hours**, especially in registration and consultation departments, to improve operational efficiency and reduce delays.
5. **Provide online appointment booking facilities**, to reduce manual workload and make healthcare services more convenient for patients.
6. **Improve coordination between hospital departments**, such as registration, consultation, laboratory, and billing sections, to ensure smooth patient movement.
7. **Strengthen crowd management practices**, particularly during peak hours, to reduce congestion and improve patient convenience in waiting areas.
8. **Use business analytics and dashboard monitoring**, to identify operational bottlenecks and support better decision-making in hospital management.
9. **Improve consultation process efficiency**, by managing doctor schedules effectively and reducing consultation waiting time.
10. **Adopt paperless healthcare systems**, to improve operational speed and reduce delays caused by manual documentation processes.
11. **Provide real-time waiting time information**, through digital displays or mobile notifications, to keep patients informed and reduce frustration.
12. **Conduct regular staff training programs**, to improve communication skills, patient handling, and operational efficiency.
13. **Collect patient feedback regularly**, to understand service-related issues and improve healthcare quality based on patient expectations.
14. **Develop separate service counters**, for inquiry, billing, and registration processes to reduce crowding and improve service speed.
15. **Continuously monitor patient waiting time and hospital performance**, using data analysis tools and reports to improve patient satisfaction and healthcare service quality.

# ANNEXURE

# QUESTIONNAIRE

- ## Section A: Basic Information

**Name (संपूर्ण नाव)**

**Gender (लिंग)**

- male
- female
- other

**Age (वय)**

- ## Section B:Patient Waiting Time Analysis

**1.Waiting time for registration was reasonable.**

- strongly agree
- agree
- neutral
- disagree
- Strongly Disagree

**2.I received timely updates about my turn.**

- Strongly agree
- Agree
- Neutral
- Disagree
- Strongly disagree

**3.The waiting time at hospital is too long**

- strongly agree
- agree
- neutal
- disagree
- strongly disagree

**4.The waiting area is comfortable**

- Strongly agree
- Agree
- Neutral
- Disagree
- Strongly disagree

**5. I am statisfied with the overall waiting time**

- strongly agree
- agree
- neutral
- disagree
- strongly disagree

- ## Sec C:To identify operational factors causing delays

**1.Do you feel staff availability affects waiting time.**

- strongly agree
- agree
- neutral
- disagree
- strongly disagree

**2.Are there delays in registration process.**

- strongly agree
- agree
- neutral
- disagree
- strongly disagree

**3.Are hospital systems (manual/digital) efficient.**

- strongly agree
- agree
- neutral
- disgaree
- strongly disagree

**4.Do you observe crowd management issues.**

- strongly agree
- agree
- neutral
- disagree
- strongly disagree

**5.Is there proper communication between staff and patients.**

- Strongly agree
- Agree
- Neutral
- Disagree
- Strongly disagree

- ## Sec D:To recommend optimisation strategies

**1.Would appointment system reduce waiting time.**

- yes
- no
- maybe

**2.Should hospital increase staff during peak hours.**

- yes
- no
- maybe

**3.Do you think digital queue system will help.**

- Yes
- No
- Maybe

**4.What improvements do you suggest?**

# REFERENCES


1. Brown, T. & Wilson, J. (2026). Healthcare Process Optimization Using Data Analytics. Oxford University Press.
2. Patel, R. (2025). "Operational Efficiency and Patient Waiting Time in Multi-Specialty Hospitals." International Journal of Healthcare Research, Vol. 14, No. 2, pp. 52–67.
3. Singh, A. & Deshmukh, P. (2025). "Application of Business Analytics in Hospital Management." Journal of Healthcare Analytics, Vol. 11, No. 4, pp. 40–55.
4. Lee, M. (2024). Digital Transformation in Healthcare Services. Springer Publications.
5. Thomas, K. & Roy, S. (2024). "Patient Satisfaction and Queue Management in Hospitals." International Journal of Hospital Administration, Vol. 9, No. 1, pp. 18–29.
6. Anderson, P. (2023). Hospital Workflow Analysis and Service Improvement. McGraw Hill Education.
7. Chavan, S. & Patwardhan, R. (2023). "Role of Technology in Reducing Hospital Waiting Time." Indian Journal of Healthcare Management, Vol. 8, No. 3, pp. 33–47.
8. World Health Organization (WHO). (2023). Improving Healthcare Service Delivery Through Technology. Geneva: WHO Publications.
9. National Health Authority, India. (2024). Digital Healthcare and Hospital Management Systems Report. Retrieved from https://www.nha.gov.in
10. Microsoft Healthcare Analytics. (2025). Data Analytics Solutions for Healthcare Operations. Retrieved from https://www.microsoft.com/en-in/industry/healthcare

# BIBLIOGRAPHY


1. Gupta, S. (2026). Advanced Healthcare Analytics and Patient Flow Management. New Delhi: Healthcare Publications.
2. Sharma, R. & Kulkarni, P. (2025). Business Analytics in Modern Hospital Systems. Mumbai: Himalaya Publishing House.
3. Jain, A. (2025). Hospital Waiting Time Optimization Using Data Analytics. Pune: Academic Healthcare Publishers.
4. Ghosh, S. (2024). Digital Queue Management in Healthcare Services. Delhi: Medical Management Publications.
5. Kumar, V. (2024). Healthcare Operations and Business Intelligence. Bangalore: Pearson Education India.
6. Mehta, R. & Shah, N. (2023). Patient Satisfaction and Healthcare Data Analysis. Chennai: McGraw Hill Education.
7. Patil, M. (2022). Hospital Workflow and Operational Efficiency. Nashik: Universal Publications.
8. Turban, E., Sharda, R., & Delen, D. (2021). Business Intelligence, Analytics, and Data Science. Pearson Education.
9. Hospital Management Journal. (2020). "Factors Affecting Patient Waiting Time and Service Quality in Hospitals." Vol. 10, No. 2, pp. 31–40.
10. Verma, K. (2019). Hospital Operations Management and Service Quality. Jaipur: Healthcare Research Publications.
11. Rao, P. (2017). Introduction to Healthcare Analytics. Hyderabad: Sage Publications.


- **Web Resources**


- World Health Organization (WHO)

https://www.who.int

- Ministry of Health and Family Welfare, Government of India

https://www.mohfw.gov.in

- Healthcare Analytics Association

https://www.healthcareanalytics.com

- IBM Business Analytics for Healthcare

https://www.ibm.com/analytics/healthcare

## Tools Used:

- **Microsoft Excel**
    - Used for data cleaning, data entry, tabulation, calculations, and chart preparation.
- **Pivot Tables and Pivot Charts**
    - Used for analyzing patient responses and creating dynamic charts for dashboard analysis.
- **Dashboard Analysis**
    - Used for visual representation of patient waiting time, satisfaction levels, and operational factors.
- **Business Analytics Techniques**
    - Used for analyzing operational inefficiencies and identifying optimization strategies.
- **Google Forms**
    - Used for collecting responses from hospital patients through questionnaires.
- **Charts and Graphs**
    - Pie charts, bar charts, doughnut charts, and column charts were used for data interpretation and presentation.
- **Observation Method**
    - Used for studying patient flow, waiting time, and hospital operational processes directly.
- **Questionnaire Method**
    - Used to collect primary data from respondents regarding waiting time and hospital services.